

LEVEL 2 REPORT

Query Intelligence

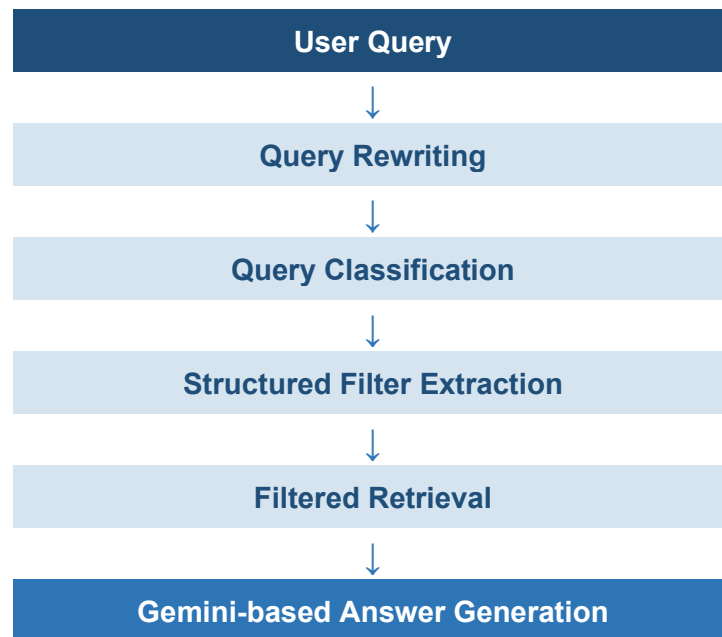
Retrieval-Augmented Generation Pipeline

Graduation Project — Technical Documentation

Objective

The objective of Level 2 is to improve the Retrieval-Augmented Generation (RAG) pipeline by introducing query intelligence before document retrieval is performed. Rather than querying the vector database directly with raw user input, the system first analyzes the query, extracts structured filtering conditions, classifies the user intent, and then performs a targeted retrieval to produce a more accurate and context-aware answer.

The updated pipeline is structured as follows:



2.1 Query Rewriting

Purpose

Users formulate queries using varied writing styles that are often short, ambiguous, or syntactically incomplete. Left unprocessed, such input reduces retrieval precision and may yield irrelevant documents from the vector database. The Query Rewriting step addresses this by transforming each raw user query into a cleaner, more retrieval-friendly formulation before any search is performed, while preserving the original intent.

The Gemini API is used to perform the rewriting. Each query is submitted to the model, which returns a grammatically complete and semantically precise version suitable for downstream retrieval.

How It Works

1. The user submits a raw query.
2. The system forwards the query to the Gemini API with a rewriting instruction.
3. Gemini returns a revised, fully-formed English query.
4. The rewritten query is passed to the next pipeline stage.

Example

Version	Query
Original	cheap five star books
Rewritten	Recommend books with a five-star rating priced under £20.

Advantages

- Removes lexical and syntactic ambiguity from user queries.
- Produces consistent, grammatically complete English input.
- Improves embedding quality, resulting in higher retrieval accuracy.
- Reduces the risk of poor matches due to informal or abbreviated phrasing.

2.2 Query Classification

Purpose

After rewriting, the system classifies the query according to its underlying intent. Understanding the query type enables the system to tailor its retrieval and response strategies more effectively. Classification is performed using a rule-based approach driven by keyword matching, which is both computationally efficient and deterministic.

Supported Categories

Category	Description	Example Query
Recommendation	User seeks book suggestions.	Recommend a good mystery book.
Comparison	User compares two or more books.	Compare Book A and Book B.
Price-related	Query involves pricing constraints.	Books under £20.
Date-related	Query references publication dates.	Books published in 2018.
Factual Lookup	User requests a specific attribute.	What is the rating of this book?
General Question	Broad informational query.	Tell me about this book.

How It Works

The classification algorithm scans the rewritten query for predefined keywords associated with each category. The first matching category is assigned. If no keyword matches, the query defaults to the General Question category.

Example

Field	Value
Input Query	Recommend five-star books under £20.
Assigned Category	Recommendation

Advantages

- Provides structured intent context before retrieval begins.
- Enables category-specific downstream processing strategies.
- Lightweight and fast — no model inference required.
- Easily extendable with additional categories and keyword rules.

2.3 Structured Filter Extraction

Purpose

Many user queries carry implicit filtering conditions that should constrain the set of candidate documents before similarity search is performed. Rather than passing these conditions as part of the query embedding, the system explicitly extracts them into a structured dictionary. This approach separates semantic retrieval from metadata filtering, improving both precision and efficiency.

Extracted Filters

Filter Field	Type	Source in Query
rating	String (e.g., "Five")	Rating keywords (e.g., "five-star")
max_price	Numeric (e.g., 20)	Price constraints (e.g., "under £20")

How It Works

- 5. The rewritten query is scanned using regular expressions and keyword matching.
- 6. Rating mentions (e.g., "five-star") are mapped to standardized string values.
- 7. Price constraints (e.g., "under £20", "less than £15") are parsed to numeric values.
- 8. Extracted values are stored in a structured dictionary passed to the retrieval stage.

Example

Input Query:

```
Recommend five-star books under £20.
```

Extracted Filters:

```
{
  "rating": "Five",
  "max_price": 20
}
```

Advantages

- Decouples metadata filtering from semantic retrieval for cleaner architecture.
- Reduces the vector search space, improving retrieval speed.
- Ensures only qualifying documents are considered, increasing answer accuracy.
- Produces machine-readable output that integrates directly with the FAISS pipeline.

2.4 Filtered Retrieval

Purpose

The Filtered Retrieval stage narrows the candidate document pool before performing similarity search against the FAISS index. By pre-filtering documents using structured metadata extracted in the previous stage, the system avoids searching across irrelevant entries and ensures that only documents satisfying all filter conditions are considered during retrieval.

Metadata Schema

Each document in the collection is associated with the following metadata fields:

Field	Type	Description
title	String	Book title.
price	String (e.g., "£18.95")	Listed price of the book.
rating	String (e.g., "Five")	Reader rating category.
source_url	String	URL of the original product page.

Metadata Example

```
{
  "title": "A Light in the Attic",
  "price": "£18.95",
  "rating": "Five",
  "source_url": "https://books.toscrape.com/catalogue/..."
}
```

Process

- Retrieve the structured filters produced in Section 2.3.
- Iterate over all stored documents and compare metadata against filter conditions.
- Retain only documents where all filter values match.
- Generate embeddings for the filtered document subset.
- Perform FAISS similarity search within the filtered subset.
- Return the Top-K most semantically relevant documents.

Advantages

- Dramatically reduces the search space for FAISS similarity computation.
- Eliminates irrelevant documents before they can influence retrieval results.
- Increases the precision of the Top-K results passed to the generation model.
- Reduces token consumption by providing Gemini with only relevant context.

2.5 Gemini-based Answer Generation

Purpose

Following filtered retrieval, the Top-K most relevant document chunks are consolidated into a structured context and submitted to the Gemini language model for answer generation. The model is instructed to produce responses grounded exclusively in the retrieved information, preventing fabrication and ensuring that all answers are traceable to the source dataset.

Prompt Structure

The prompt submitted to Gemini contains three components:

Prompt Component	Content
Retrieved Context	Concatenated text chunks from the Top-K retrieved documents.
User Question	The original rewritten query from Section 2.1.
Instruction	A directive to answer only using the provided context.

Workflow

15. Top-K document chunks are concatenated into a single context string.
16. The context, user query, and generation instruction are assembled into a prompt.
17. The prompt is submitted to the Gemini API.
18. Gemini generates a response grounded in the retrieved context.
19. The response is returned to the user.

Advantages

- Grounds all generated answers in verified, retrieved documents.
- Prevents hallucinations by explicitly restricting the model to the provided context.
- Produces coherent, natural-language responses from structured book data.
- Leverages Gemini's instruction-following capabilities for reliable, reproducible output.

Conclusion

Level 2 introduces a multi-stage query intelligence layer that significantly enhances the capabilities of the RAG pipeline established in Level 1. By processing each user query through rewriting, classification, and structured filter extraction before retrieval, the system acquires a precise understanding of user intent prior to any database interaction.

The filtered retrieval mechanism ensures that FAISS similarity search operates only over documents satisfying explicit metadata constraints, resulting in higher precision and reduced computational overhead. Gemini-based answer generation, constrained to the retrieved context, produces reliable, hallucination-free responses directly grounded in the dataset.

Together, these components transform the system from a basic semantic search tool into an intent-aware retrieval engine. The structured pipeline established in Level 2 provides the architectural foundation required for more advanced capabilities — such as multi-query expansion, re-ranking, and conversational memory — in subsequent levels.

Component	Benefit Delivered
Query Rewriting	Improves embedding quality and retrieval accuracy.
Query Classification	Enables intent-aware processing strategies.
Filter Extraction	Reduces search space; enforces metadata constraints.
Filtered Retrieval	Increases precision of Top-K results.
Answer Generation	Produces grounded, hallucination-free responses.